

Economic History II

Institutions and the reversal of fortune

FGV EPGE

What is this course?

A **topics** course in economic history.

Each class uses an empirical paper as the entry point to a larger question:

- Did the slave trade reduce interpersonal trust in African societies today?
- Did Jesuit missions among the Guarani raise education and income 250 years later?
- Did reforming the postal service and reducing patronage raise state capacity and innovation in the US?
- Did banning German in American schools after the First World War integrate immigrants or generate backlash?

How historical evidence can be used to study economics.

Why study economic history?

- Many present-day differences in income, trust, state capacity, institutions and political identity have historical roots.
- The past offers variation that we would not create by experiment.
- Economic history combines theory, data and context to distinguish causation from correlation.

- **Topics:** a collection of themes. What the papers have in common are the methods, the tools and the format. They do not share a theme, a period or a region. And there is no chronological order.
- **History:** the past.
- **Economic:**
 - Economics is the study of how society manages its scarce resources (Mankiw, 2001, p. 4).
 - Economics is the social science that studies the choices that individuals, businesses, governments and entire societies make as they cope with scarcity (Bade and Parkin, 2002, p. 5).
 - Economics is the study of human behaviour, with a focus on decision making (Gwartney et al., 2006, p. 5).
 - Economics is what economists do (Jacob Viner?).
- Why so much politics?

Course themes

- Institutions and comparative development.
- Causality and methods in economic history.
- Colonialism and historical persistence.
- Deep origins: agriculture, cities and the state.
- Leadership, elites and state performance.
- Bureaucracy and state capacity.
- Religion, reform and public goods.
- Nation building, political identity and propaganda.

What will we do in this course?

- Read papers:
 - question;
 - data;
 - identification;
 - results;
 - mechanisms.
- Discuss what the historical evidence implies for broader economic questions.

Logistics: assessment

- 25% participation.
- 30% midterm exam.
- 10% student presentation.
- 35% final exam.

Participation

- At the start of class, each student hands in a physical sheet with three items:
 - the research question of the paper;
 - the main argument;
 - a question or comment.
 - up to 200 words, about 1/4 of a page.
- I collect the sheets at the beginning of class.
- They also serve as a proxy for attendance.

Participation: dividing the class

- The class was randomly drawn and split into two groups.
- Half hand in the sheet in the Tuesday classes.
- Half hand in the sheet in the Thursday classes.
- **25% of the grade**

Exams, presentations and support

- There will be practice versions of the midterm (**30% of the grade**) and of the final exam on the website (**35% of the grade**).
- Presentations will be in groups: 30 minutes to present a paper assigned by us (**10% of the grade**).
- **Alexandre Portugal: teaching assistant**, PhD student at FGV EPGE, works on economic history, in particular colonial Brazil.

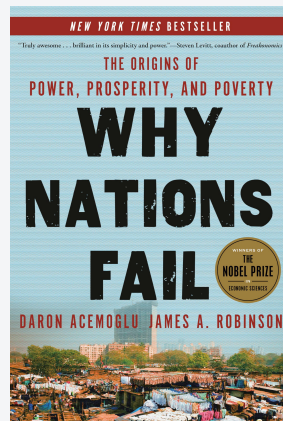
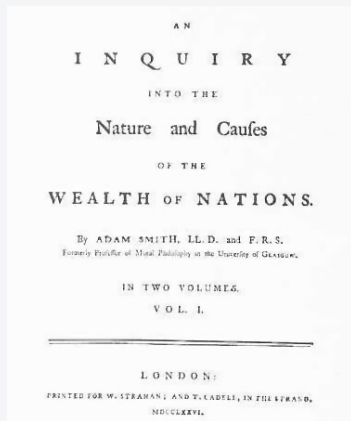
The last four Nobel prizes: a reminder

- **2022** Ben Bernanke Douglas Diamond, Philip Dybvig
- **2023** Claudia Goldin
- **2024** **Daron Acemoglu, Simon Johnson, James Robinson**
- **2025** Joel Mokyr Philippe Aghion, Peter Howitt

Today: using AJR as the entry point to institutions and comparative development.

From Smith to Acemoglu

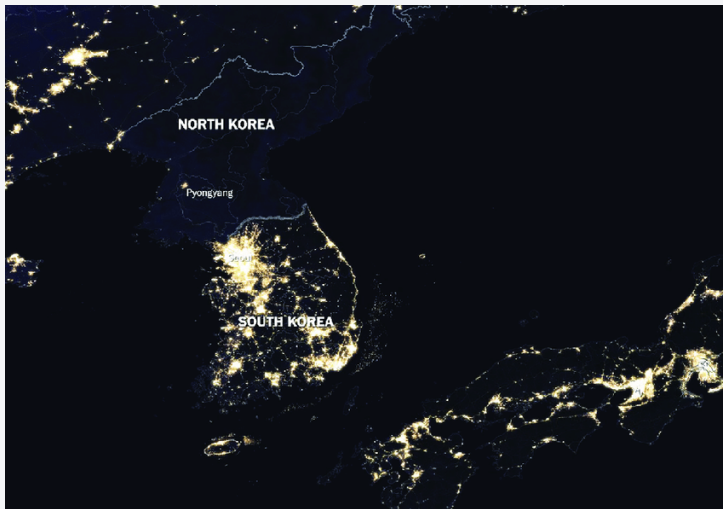
Why are some nations rich and others poor?



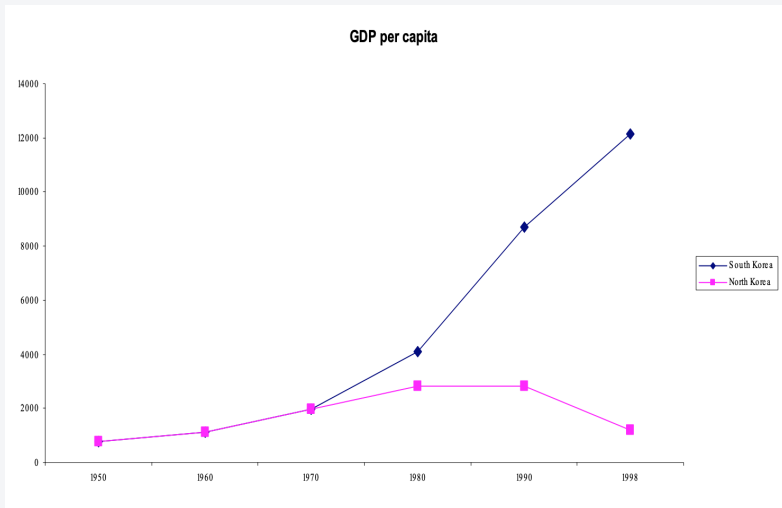
Why are some countries richer than others?

- A great question in economics, ever since the *Wealth of Nations*.
- What causes economic growth?
 - Solow (1956): capital accumulation.
 - Romer (1990): innovation.
 - Education.
 - Economies of scale.
- North and Thomas (1973): *the factors we have listed... are not causes of growth; they are growth.*
- **Institutions?** Culture? Geography?

Do institutions matter?



Do institutions matter?



What are institutions?

- In the standard microeconomic setting: agents, preferences and **constraints**.
- *Institutions are the rules of the game in a society; more formally, they are the humanly devised constraints that shape human interaction.*
- *As a consequence, they structure incentives in human exchange, whether political, social or economic. (North 1990)*

Types of institutions

- Economic institutions:
 - Property rights.
 - Market structure.
 - Land tenure, patent law, antitrust regulation, quality standards and so on.
- Political institutions:
 - Rules that regulate the distribution of political power.
 - Who takes decisions that are, ultimately, backed by force?
 - Democracy vs. dictatorship, electoral laws, eligibility, succession and so on.
- Formal vs. informal institutions.

Framework: Acemoglu, Johnson and Robinson (2005)

1. Economic institutions matter for growth.

- $\text{economic institutions}_t \rightarrow \begin{cases} \text{distribution of resources}_t \\ \text{economic performance}_t \end{cases}$

2. Economic institutions are endogenous.

- $\text{political power}_t \rightarrow \text{economic institutions}_t$

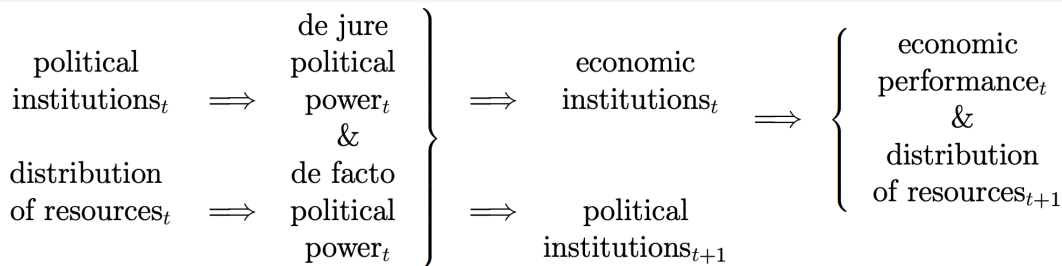
3. Political power is also endogenous.

- $\text{political institutions}_t \rightarrow \text{political power}_t$

Framework: Acemoglu, Johnson and Robinson (2005)

4. Political power is also endogenous, part 2.
 - distribution of resources _{t} $\rightarrow$ political power _{t}
5. Dynamic political institutions:
 - political power _{t} $\rightarrow$ political institutions _{$t+1$}
6. Commitment problems.

Framework: Acemoglu, Johnson and Robinson (2005)



What are **good** institutions?

Acemoglu, Johnson and Robinson (2005)

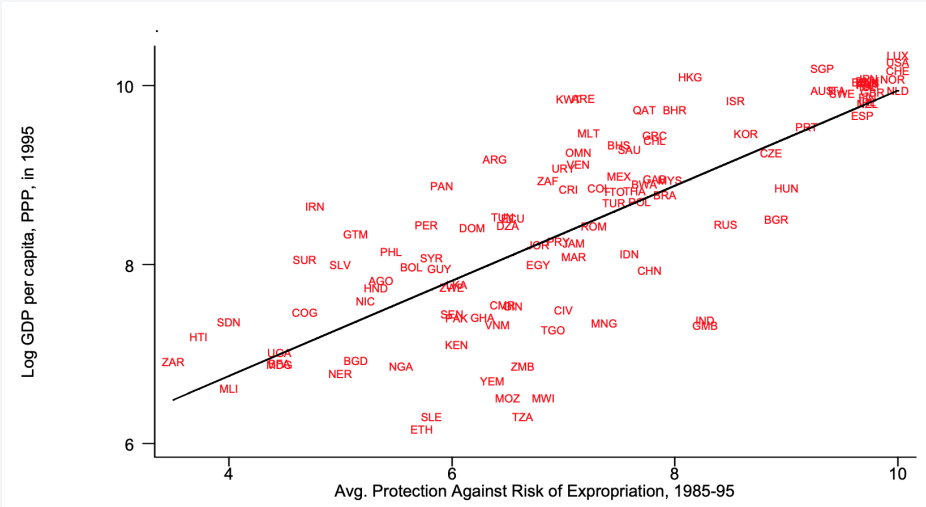
- *Good economic institutions are those that provide security of property rights and relatively equal access to economic resources to a broad cross-section of society.*
- Security of property rights → investment.
- Broad access → more people can produce and innovate.
- It is not clear, in practice, which institutions do this.
- There may be conflicts between these objectives.

Some corollaries

- Institutions that limit political power reduce expropriation.
- Good economic institutions tend to emerge when political power is dispersed.
- They also tend to emerge when those who hold power have no large rents to expropriate.
- Power begets power: a high degree of historical persistence.

Do institutions matter?

- Economic:
 - What are they?
 - How do we measure them?
 - How do we establish a causal relationship between economic institutions and development?
- Political institutions: harder still.



Reversal of Fortune

Acemoglu, Johnson and Robinson (2002, QJE)

Question

- In 1500, some regions that would later be colonized were relatively prosperous: dense populations, large cities, complex political organization.
- By the end of the twentieth century, many of those same regions were relatively poor.
- AJR call this the **reversal of fortune**.
- The explanation: European colonialism changed the institutional path of initially prosperous places.

Two measures of prosperity in 1500

- **Urbanization**: cities require an agricultural surplus, trade, transport and political organization.
- **Population density**: prosperous regions can sustain larger populations.
- Both measures are imperfect. AJR check whether the pattern holds with each of them.
- The sample of interest consists of former European colonies.

TABLE I
DESCRIPTIVE STATISTICS

	Whole world (1)	Base sample for urbanization (2)	Base sample for population density (3)	Below median urbanization in 1500 (4)	Above median urbanization in 1500 (5)	Below median population density in 1500 (6)	Above median population density in 1500 (7)
Log GDP per capita (PPP) in 1995	8.3 (1.1)	8.5 (0.9)	7.9 (1.0)	8.8	8.1	8.3	7.5
Urbanization in 1995	53.0 (23.8)	57.5 (22.4)	45.4 (22.2)	64.9	49.7	53.5	36.7
Urbanization in 1500	7.3 (5.0)	6.4 (5.0)	6.4 (5.0)	2.4	10.5	2.3	9.5
Log population density in 1500	1.0 (1.6)	0.2 (1.9)	0.5 (1.5)	-0.9	1.4	-0.6	1.6
Population density in 1500	9.2 (24.3)	6.3 (16.4)	4.8 (11.7)	1.2	11.7	0.8	9.1
Log population density in 1000	0.6 (1.5)	0.11 (2.0)	0.08 (1.5)	-1.20	1.22	-0.94	1.04
Average protection against expropriation, 1985-1995	7.1 (1.8)	6.9 (1.5)	6.5 (1.4)	7.5	6.3	6.8	6.2
Constraint on the executive in 1990	3.6 (2.3)	4.9 (2.1)	3.7 (2.3)	5.1	4.6	4.0	3.5
Constraint on the executive in first year of independence	3.6 (2.4)	3.3 (2.5)	3.4 (2.3)	3.8	2.8	3.6	3.3
European settlements in 1900	29.6 (41.7)	23.2 (28.7)	12.5 (22.1)	30.5	6.0	18.7	4.7
Number of observations	162	41	91	21	20	47	44

Table I

- Descriptive statistics split former colonies by prosperity in 1500.
- Above-median urbanization in 1500: lower log GDP in 1995, less protection against expropriation, less European settlement in 1900.
- Above-median population density in 1500 shows the same pattern.
- The regions that were most prosperous in 1500 have lower income and weaker institutions today.

TABLE II
URBANIZATION AND PER CAPITA INCOME

	Cross-sectional regression in 1913, all countries (1)	Cross-sectional regression in 1950, all countries (2)	Cross-sectional regression in 1995, all countries (3)	Cross-sectional regression in 1995, only for ex-colonies (4)	Cross-sectional regression in 1995, never colonized countries only (5)	Cross-sectional regression in 1995, all countries, with continent dummies (6)	Panel data set through 1913 (7)
<i>Dependent variable is log GDP per capita</i>							
Urbanization	0.038 (0.006)	0.026 (0.002)	0.036 (0.002)	0.037 (0.003)	0.033 (0.007)	0.030 (0.002)	0.026 (0.004)
R^2	0.69	0.57	0.63	0.69	0.34	0.68	0.93
Number of observations	22	128	162	93	51	162	55

Table II

- The paper first establishes that urbanization is a reasonable proxy for income.
- Across countries, urbanization is positively associated with log GDP per capita in 1913, 1950 and 1995.
- The relationship holds for all countries, for former colonies only, for never-colonized countries only, and with continent dummies.
- Figures I and II reverse this relationship when prosperity is measured in 1500.

Log GDP per capita, PPP, 1995

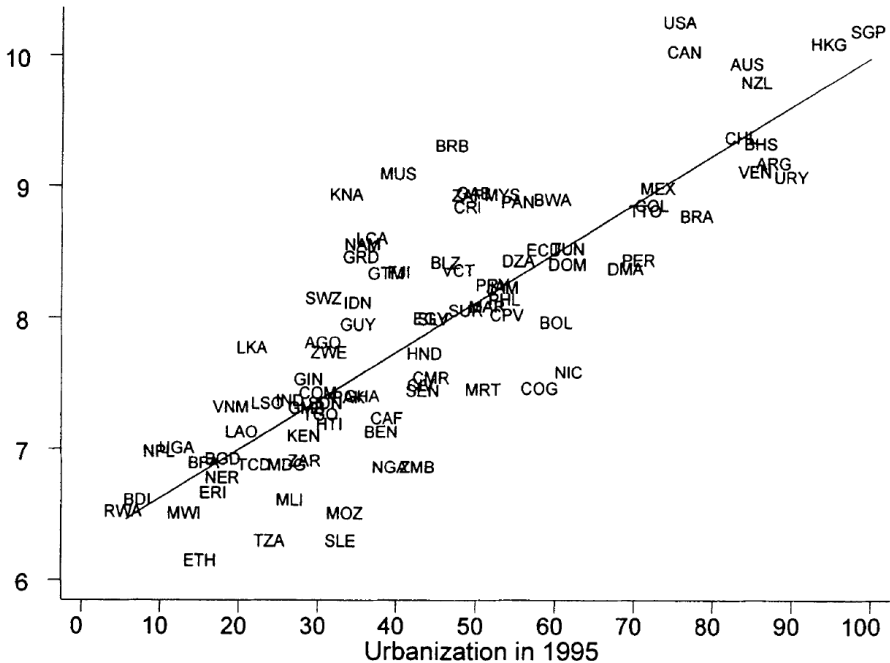


Figure III

- In the modern cross-section, urbanization and income move together.
- So the negative relationship in 1500 is not mechanical.
- The paper asks why the sign flips among former colonies.

Log GDP per capita, PPP, 1995

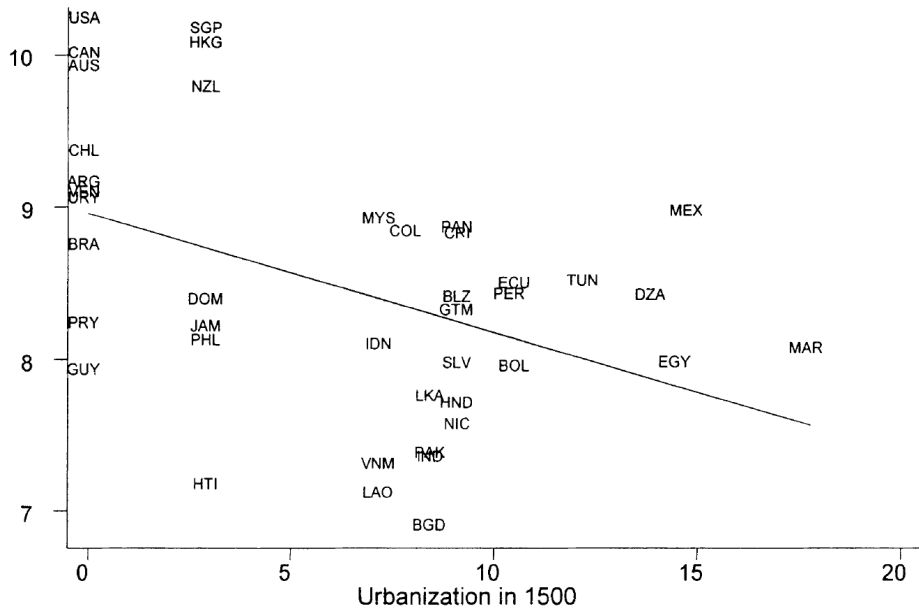


Figure I

- Countries with higher urbanization in 1500 tend to have lower income in 1995.
- The group that was prosperous in 1500 includes Mexico, Peru, Egypt and India.
- The group that was not prosperous in 1500 includes the United States, Canada, Australia and New Zealand.

Log GDP per capita, PPP, 1995

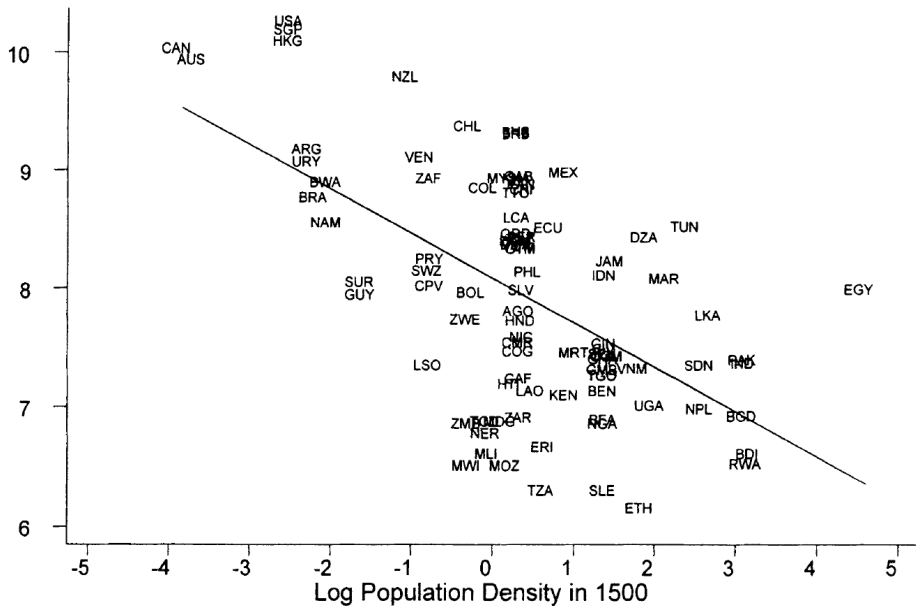


Figure II

- Population density in 1500 produces the same qualitative result.
- Density covers countries with no urbanization data.
- It also brings sub-Saharan Africa into the analysis, where data on city sizes are weaker.

Main regression evidence

- Table III regresses log GDP per capita in 1995 on urbanization in 1500 for former colonies.
- Baseline coefficient: about -0.078 (standard error 0.026).
- The result survives dropping North Africa, dropping the neo-Europes and adding continent dummies.
- It is also robust to latitude, temperature, humidity, access to the sea, resources, colonial power and religion.

Population density and robustness

- Table IV uses alternative urbanization series: Bairoch, Eggimann, Chandler and the Davis-Zipf adjustment.
- Table V repeats the result with log population density in 1500.
- Baseline coefficient for density: about -0.38 ; density 10% higher in 1500 predicts current income about 4% lower.
- Table VI uses population density as an instrument for urbanization; the reversal remains.

Measurement problems

- The 1500 data are noisy.
- Classical measurement error should attenuate the coefficients towards zero.
- That is why AJR put weight on similar results using urbanization, population density, alternative series and IV.
- AJR emphasize the robustness of the sign of the reversal, rather than the precision of the coefficient.

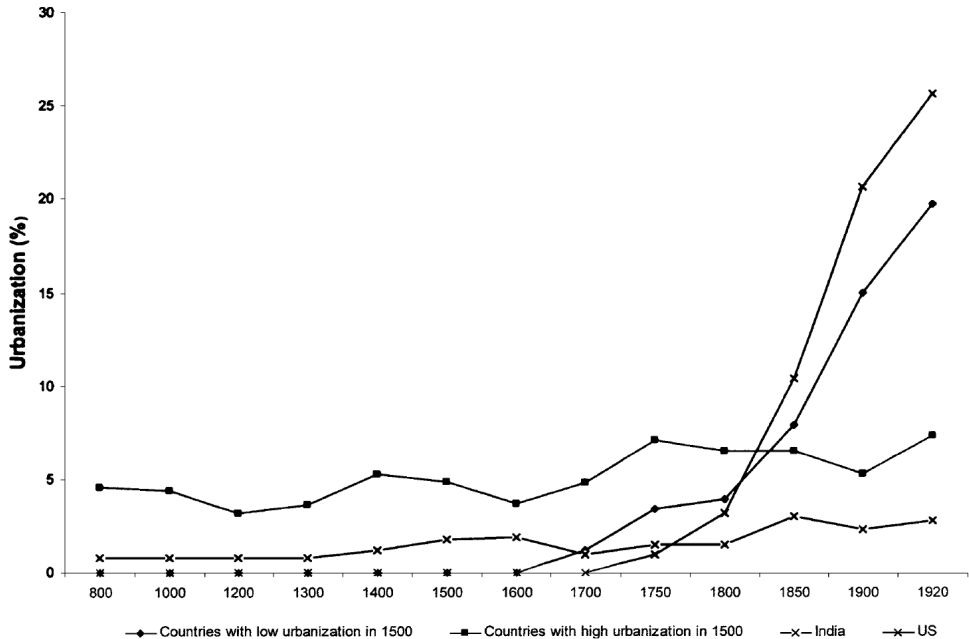


Figure IVa

- The initially less urbanized New World group catches up with and overtakes the initially highly urbanized group.
- The key moment is between 1750 and 1850.
- India and the United States make the contrast especially visible.

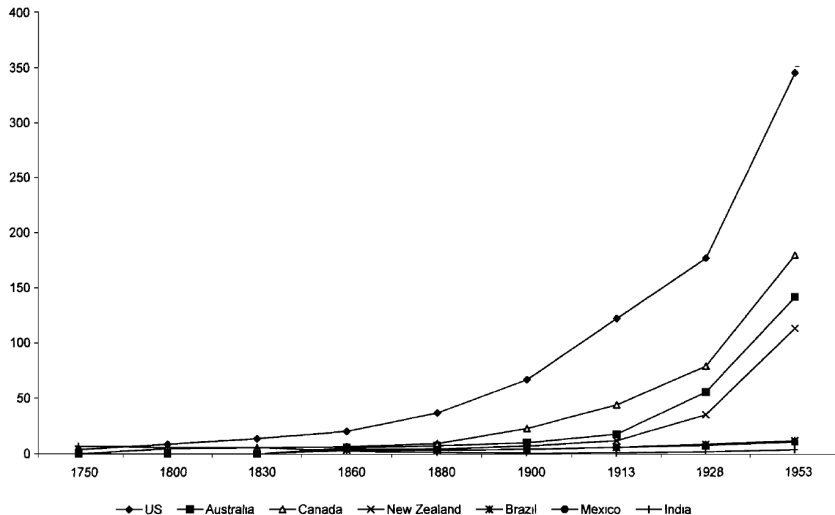


FIGURE IVb

Industrial Production per Capita, 1750–1953

Note. Index of industrial production with U. K. per capita industrialization in 1900 is equal to 100, from Bairoch [1982].

Figure IVb

- The reversal coincides with industrialization.
- The United States, Canada, Australia and New Zealand take off relative to Brazil, Mexico and India.
- This timing is hard to reconcile with explanations that would operate immediately after the conquest.

Against simple geography

- A fixed-geography story predicts persistence: places favoured in 1500 should have remained favoured.
- The reversal goes in the opposite direction.
- AJR also argue that variants based on agricultural technology, disease, coal or access to the sea do not fit the timing and the robustness of the results.

The institutional hypothesis

- Europeans were more likely to create extractive institutions in densely populated and prosperous areas.
- Existing populations, tribute systems and labour coercion made extraction profitable.
- Sparsely populated areas were more likely to attract European settlement and institutions protecting property rights.
- The paper calls this the **institutional reversal**.

Evidence of institutional reversal

- Current measures of institutions are worse in places that were more prosperous in 1500.
- European settlement in 1900 is lower in initially dense and urbanized regions.
- The income reversal becomes much weaker when the regressions control for current institutions.
- Interpretation: initial prosperity affected later income through colonial institutions.

Institutions and industrialization

- Why does the reversal appear during industrialization?
- AJR argue that good institutions matter more when new investment opportunities arise.
- Industrialization raised the return to broad investment, entry and innovation.
- Extractive institutions limited that response where colonialism concentrated power.

Final message

- The paper documents a reversal in relative income among former colonies.
- The reversal appears both with urbanization and with population density in 1500.
- It happens mainly with industrialization, not immediately after colonization.
- AJR's central claim: colonial institutional choices turned initial prosperity into a long-run disadvantage.

Back to the paper: Reversal of Fortune

- **Question:** why are some regions that were rich in 1500 relatively poor today?
- **Data:** urbanization and population density before colonization; income per capita today.
- **Result:** there is a reversal of fortune between 1500 and the present.
- **Mechanism:** European colonization led to different institutions in different places.
- **Interpretation:** institutions help explain long-run economic paths.
- **What was left out:** the technical details of the regressions.